

Exam Date & Time: 08-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech V Semester - Make up Examination, Jan - 2025
ADVANCED COMPUTER NETWORKS [CSE-5151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast the challenges and issues in UAV network. (4)
 - A)
 - B) Explain the multi-UAV network for the case of COVID-19. (4)
 - C) Summarize the applications of UAVs. (2)
- 2) Explain the architecture of Software Defined Network (SDN) with a neat diagram. (4)
 - A)
 - B) Explain the working of Cisco-ACI APIC controller. (4)
 - C) With suitable example explain the network virtualization components. (2)
- 3) Discuss the components ACI in designing as an open system. (4)
 - A)
 - B) Explain the relationship between SDS and HCI. (4)
 - C) Compare block and file storage. (2)
- 4) Interpret the various types of VoIP playout delays with suitable examples. (4)
 - A)
 - B) Explain step by step procedure connect where jim@umass.edu calls keith@poly.edu for SIP. (4)
 - C) Illustrate how in Skype networks peers relay provides solution to the problem of "NAT prevents outside peer from initiating connection to insider peer". (2)

- 5) Assuming light signal of power P_t in watts (W), or milliwatts (mW). Compute the attenuation loss and amplification for the given values $(P_t)_{dBW} = 10 \log(P_t)W$ and $(P_t)_{dBm} = 10 \log(P_t)mW$. (4)
- A)
- B) Interpret different types of optical regeneration techniques. (4)
- C) What is the principle on which Optical Fiber Network works and explain the features of optical amplifiers. (2)

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